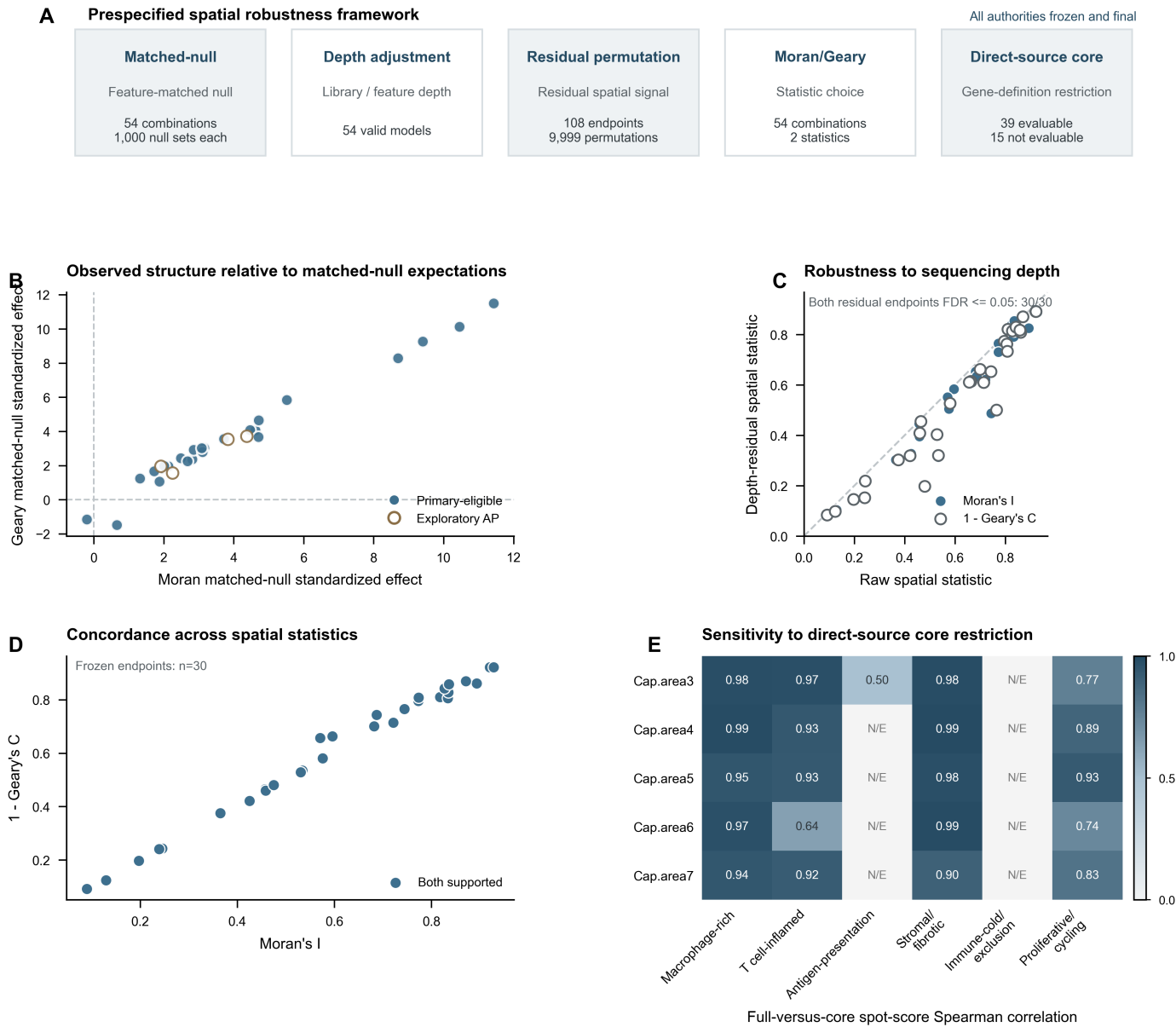


Supplementary Figure S5 | Distinct spatial sensitivity analyses and their denominators

Supplementary Fig. S5 | Robustness and sensitivity analyses of spatial continuous-program structure



Separate frozen sensitivity analyses and matched-null denominators

Matched-null support at both endpoints: 42/54 evaluable combinations overall; 25/32 primary + exploratory combinations.

Important unsupported combinations: Cap.area3 x Antigen-presentation (coverage 21/22; Moran $z=-0.201$; empirical $P=0.593$); Cap.area5 x Stromal/fibrotic; Cap.area7 x Stromal/fibrotic (both $FDR > 0.05$); and Cap.area7 x Immune-cold/exclusion.

Failures occur at both 21/22 and 22/22 coverage and are not described as coverage-only effects.

Depth-adjusted support (30/30 DLBCL-area combinations at both endpoints) is a separate analysis and does not rescue matched-null failures.